

HOMEWORK 1
due Wed, Sep. 9, 11:59 PM
(Upload your solutions to Gradescope)

1. (2 points) prove that a norm on a vector space V satisfies

$$|\|u\| - \|v\|| \leq \|u - v\| \text{ for all } u, v \in V.$$

Do not assume the norm comes from an inner product.

2. (4 points) (#1 p. 34.) Verify that the function

$$\langle Z, W \rangle = \sum_{j=1}^n Z_j \overline{W_j}$$

for $Z = (Z_1, \dots, Z_n)$, $W = (W_1, \dots, W_n) \in \mathbb{C}^n$ defines an inner product on $\mathbb{C}^n$ (i.e., satisfies Definition 0.1).

3. (2 points) (#3 p. 35.) Define $\langle V, W \rangle$ for $V = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ and $W = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \in \mathbb{C}^2$ as

$$(\overline{w_1}, \overline{w_2}) \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

Show that $\langle V, V \rangle = 0$ for all vectors $V = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ with $v_1 + 2v_2 = 0$. Does $\langle \cdot, \cdot \rangle$ define an inner product?

4. (4 points) (#9 p. 35. Find an orthonormal basis for the orthogonal complement.)

Compute the orthogonal complement of the subspace in $\mathbb{R}^3$ spanned by the vector $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$.

5. (2 points) (#10 p. 35.) Let $f(t) = 1$ on $0 \leq t \leq 1$. Show that the orthogonal complement of f in $L^2[0, 1]$ is the set of all functions whose average value is zero.

6. (2 points) (#11 p. 35.) Show that if a differentiable function, $f(t)$, is orthogonal to $\cos(t)$ in $L^2[0, \pi]$, then $f'(t)$ is orthogonal to $\sin(t)$ in $L^2[0, \pi]$. *Hint: integrate by parts.*

7. (2 points) (#17 p. 36.) Suppose u_0 and u_1 are vectors in the inner product space V with $\langle u_0, v \rangle = \langle u_1, v \rangle$ for all $v \in V$. Show that $u_0 = u_1$. *Hint: Let $v = u_0 - u_1$.*

8. (2 points) (#23 p. 37.) Show that a set of orthonormal vectors is linearly independent.

Page numbers refer to “A First Course in Wavelets with Fourier Analysis, 2nd Edition” by Boggess and Narcowich.